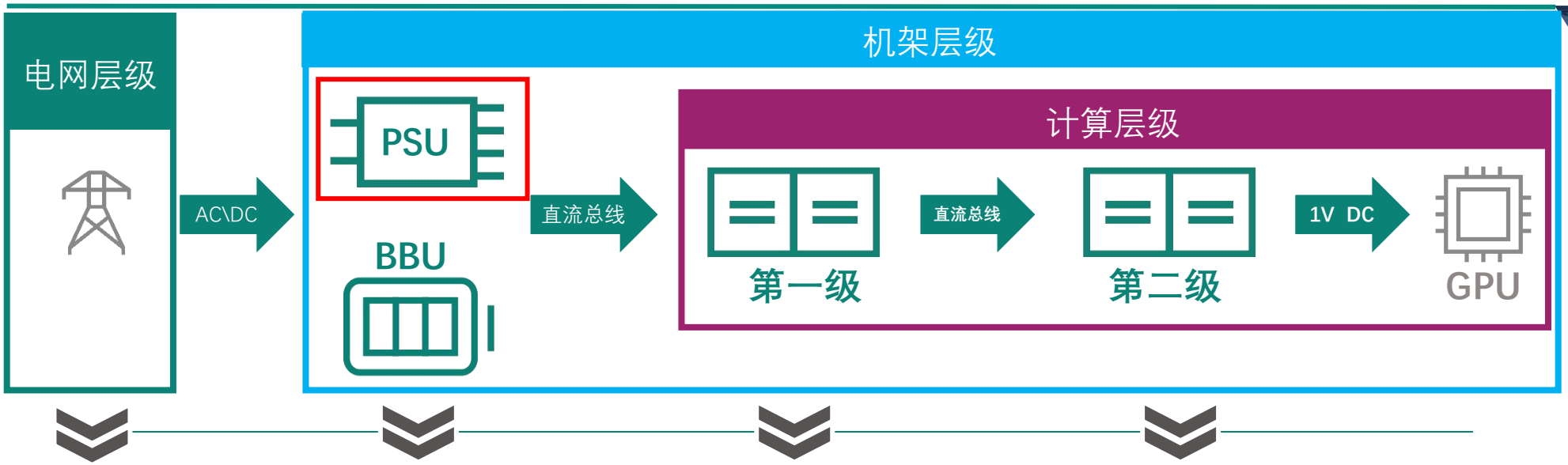
英飞凌为AI服务器提供高效率/高功率密度供电方案 – PSU 部分

2025年3月14日 深圳
2025年3月18日 北京

人工智能数据中心三个层级的电力传输趋势



要点总结

- AI 数据中心电力传输的演进速度与 GPU/TPU 的演进速度相当。
- AI 数据中心的电力需求增长速度与 GPU/TPU 的功耗增长速度相当。
- 1MW/机架的目标加速了±400V以上直流总线的普及，推动了WBG需求增长。

We power AI from Grid to Core

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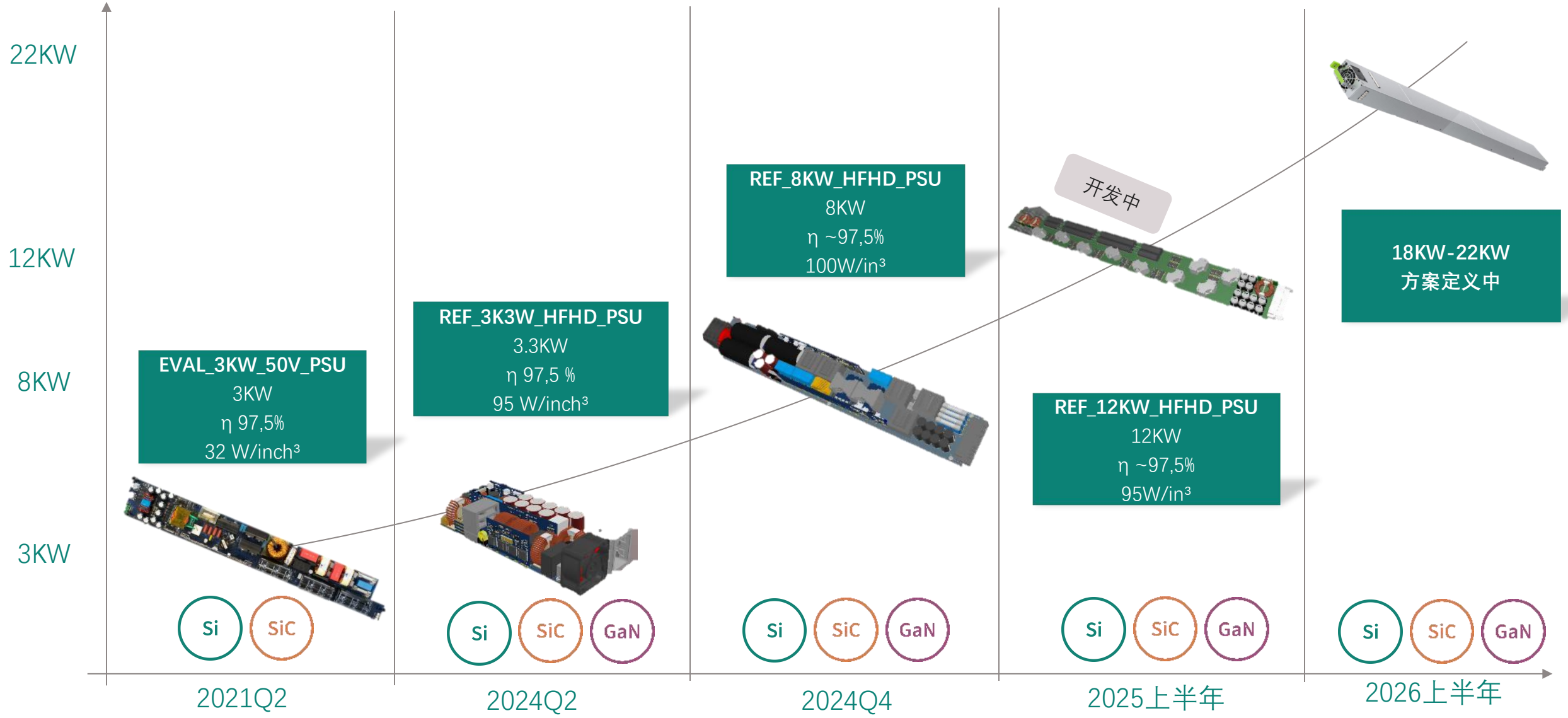
英飞凌PSU方案紧跟AI服务器供电需求演进趋势



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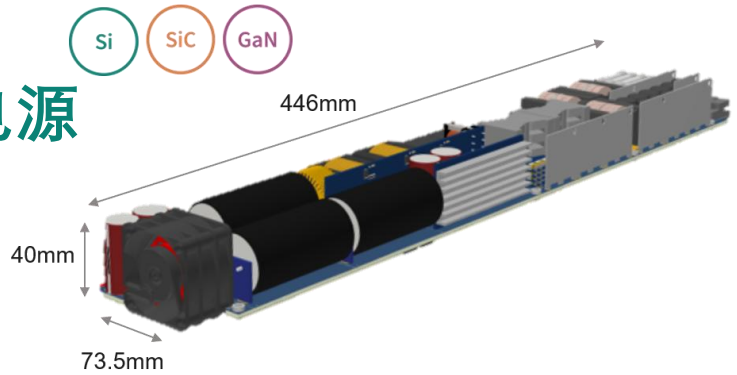
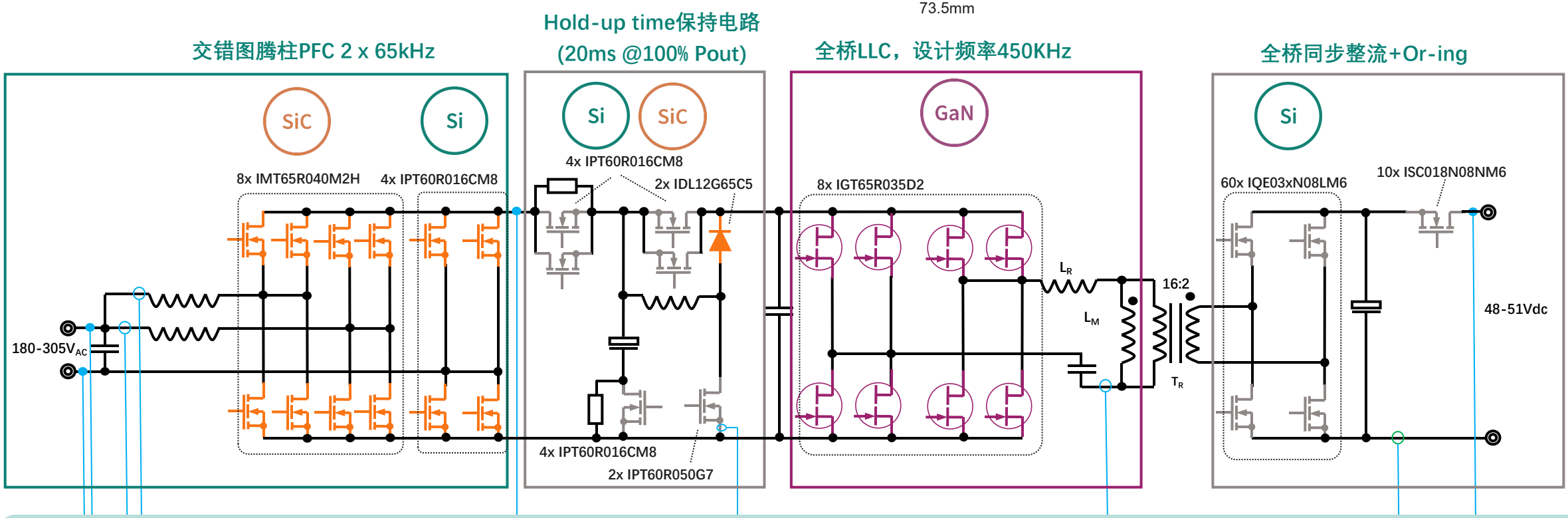
基于Si/SiC/GaN的8KW混合方案
超越ORv3标准的高效率/高功率密度AI服务器电源





DSP/MCU控制单元: PSOC™ C3

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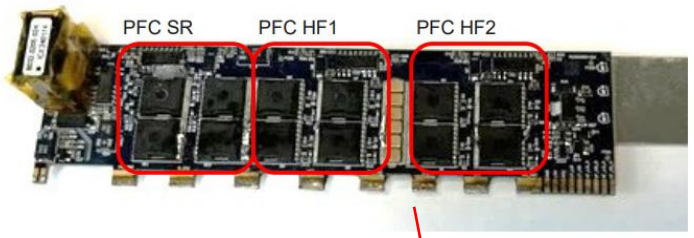
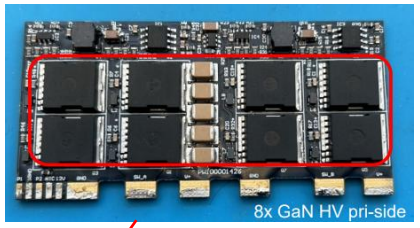
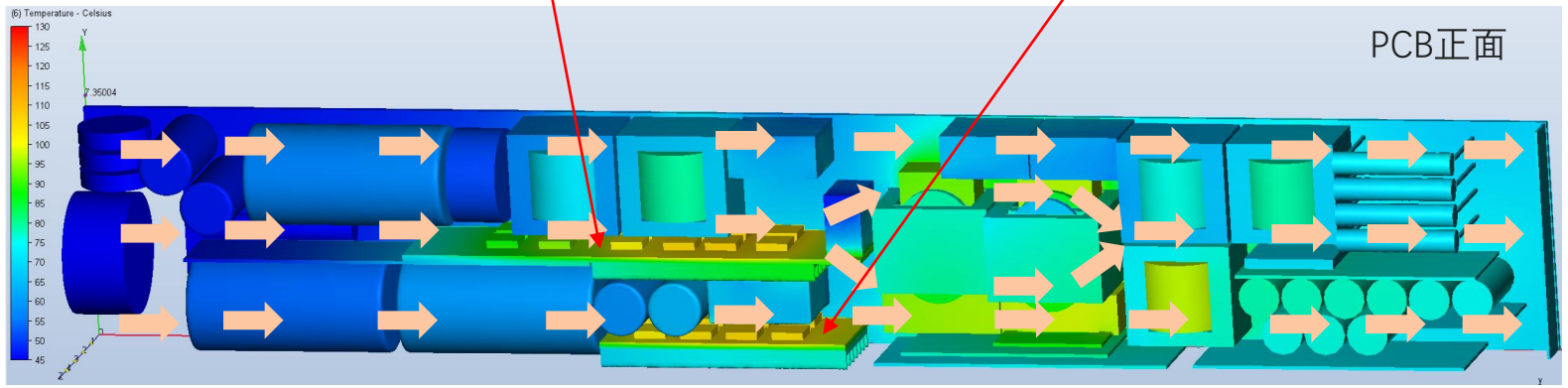
功率器件插卡式设计，有利于系统散热，提高功率密度

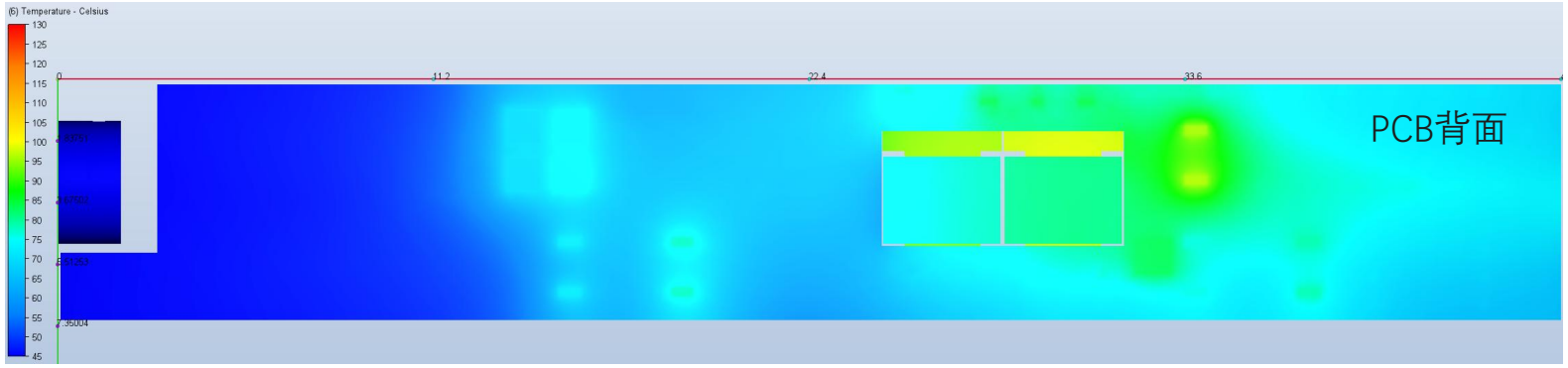
PFC插卡



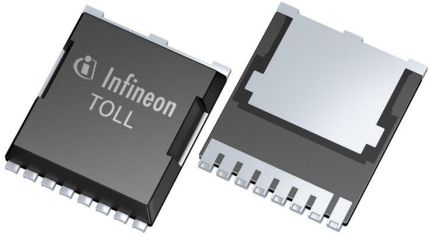
LLC插卡







TOLL封装，特别适合插卡式设计，是AI服务器电源设计首选。



Vin=180Vac, 8KW满载输出
环境温度45℃，带壳测试温度最高点

- PFC管子：107℃
- LLC管子：106℃
- 同步整流MOS: 99℃
- 变压器磁芯：80℃
- 变压器绕组：99℃

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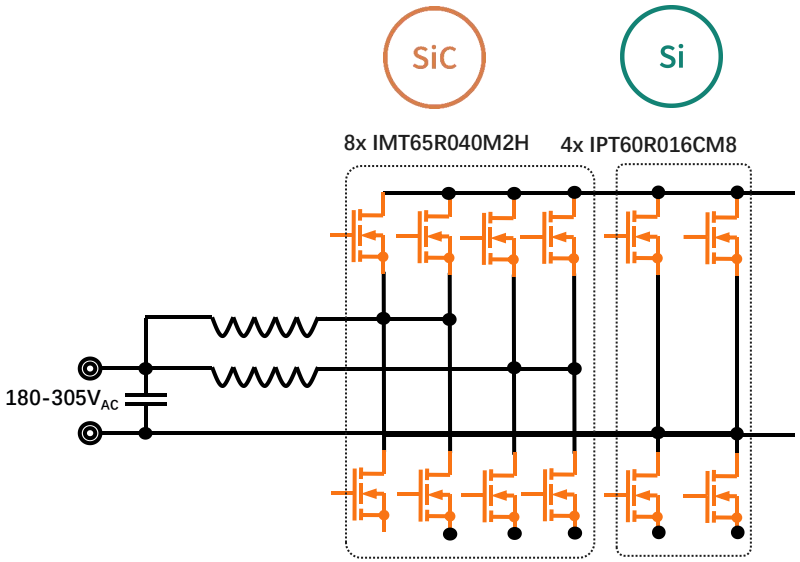
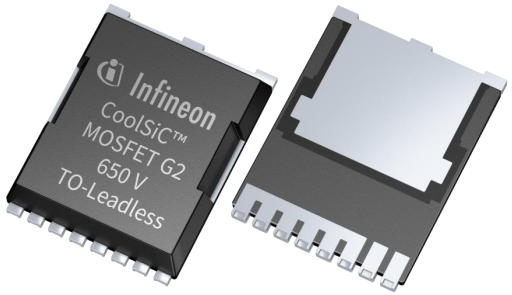
CoolSiC™ Gen2和CoolMOS™ 8
帮助交错式图腾柱PFC级实现高达99.05%的效率

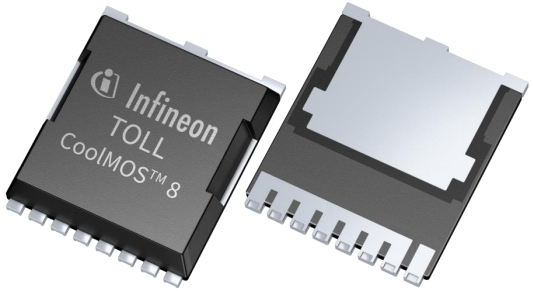
交错图腾柱PFC 2 x 65kHz





CoolSiC™ Gen2

- 支持零电压或者负电压关断，方便兼容设计
- 秉承英飞凌SiC产品一贯可靠的GOX（门极氧化层）设计，稳定可靠
- 良好的温度特性和抗浪涌能力
- 比Gen1相比，关键性能指标FOM提升30%~50%，实现更高的效率



CoolMOS™ 8

- 英飞凌最新一代CoolMOS™，最低内阻低至7mohm
- 进一步降低开关损耗，持续引领行业效率标杆
- 统一以CM8结尾，简化产品系列，更易使用
- 更具成本优势的CoolMOS™

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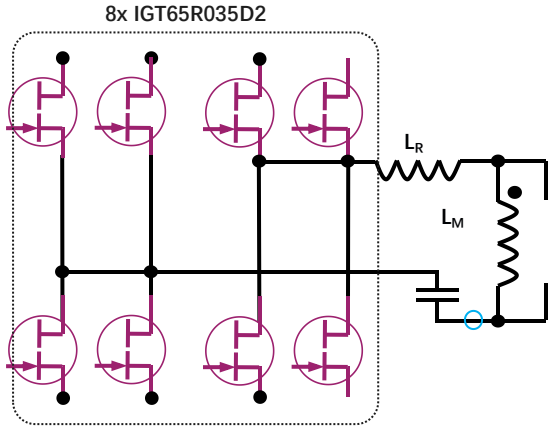
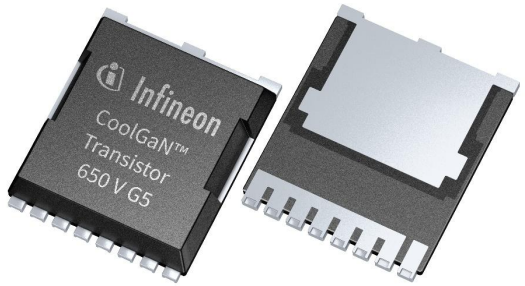
采用CoolGaN™ G5，实现全桥LLC高频化，提高功率密度

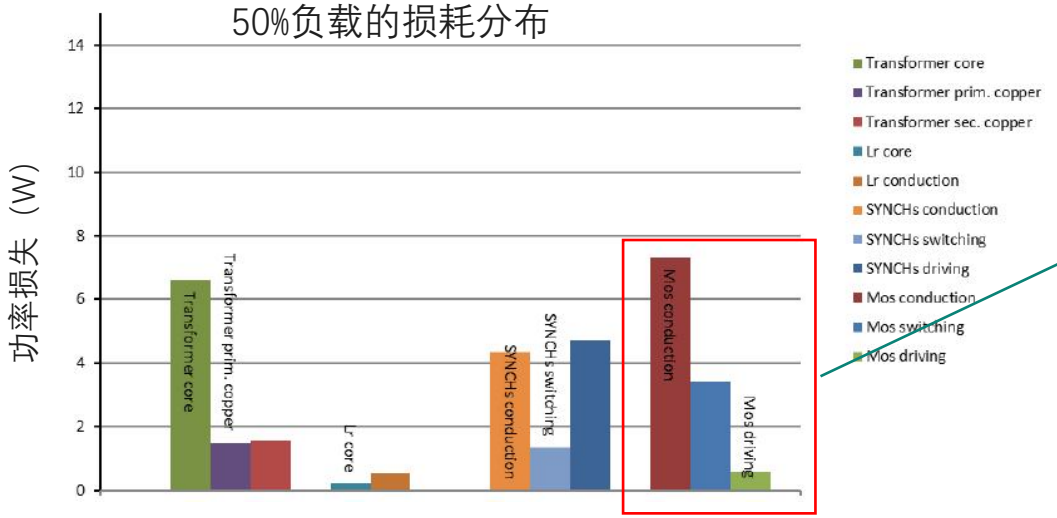
全桥LLC，工作频率350KHz~1.5MHz，
效率高达98.7%@50%负载





CoolGaN™ G5

- 采用GIT技术的E-Mode GaN，驱动更容易，抗干扰能力更强
- 与前一代产品相比，Eoss和Qoss 随Vds的变化降低约20%，大幅度降低开关损耗
- CoolGaN™ G5 提供更多的封装，以及集成驱动/保护/检测功能的氮化镓器件供不同设计选用



GaN器件的**导通**损耗，与负载大小强相关

GaN器件的**开关**损耗占比较低，体现了CoolGaN™优秀的高频工作特性

GaN器件的**驱动**损耗，体现了GIT技术对驱动功耗的需求极低

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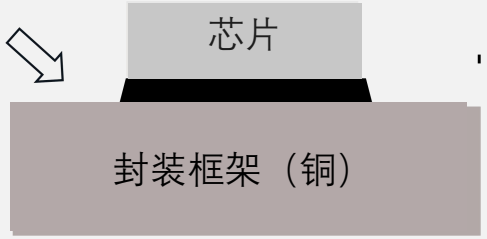
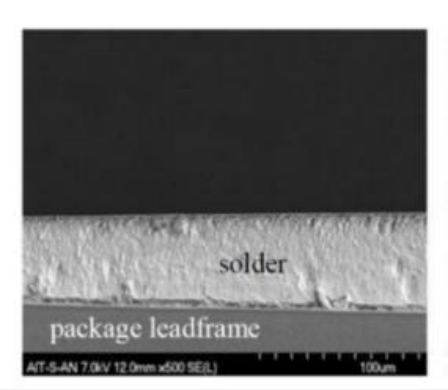
英飞凌创新的.XT熔合技术，大幅度降低器件热阻Rth.jc

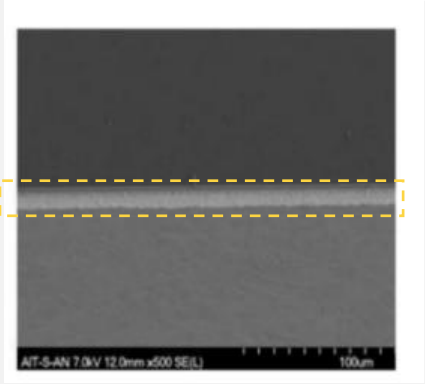
8KW方案中的CoolMOS™ 8， CoolSiC™ G2 和 CoolGaN™ G5，均采用.XT技术。

传统的焊接方式



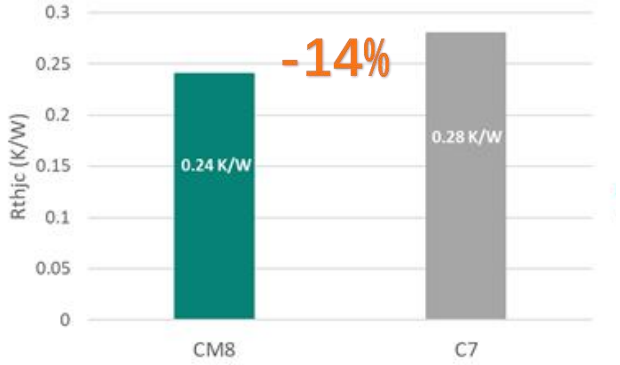


英飞凌创新的.XT熔合技术

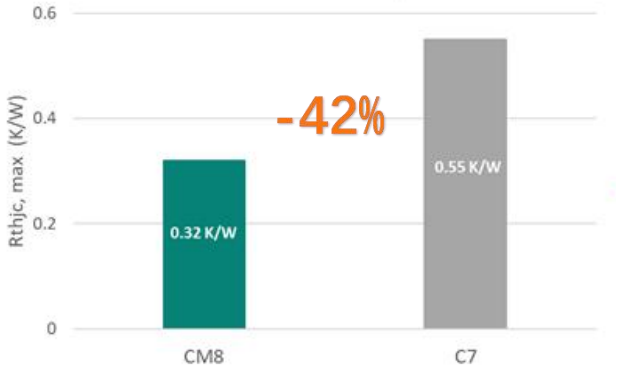


.XT技术热性能的提升

IPW60R016CM8 vs IPW60R017C7



IPP60R037CM8 vs IPP60R040C7



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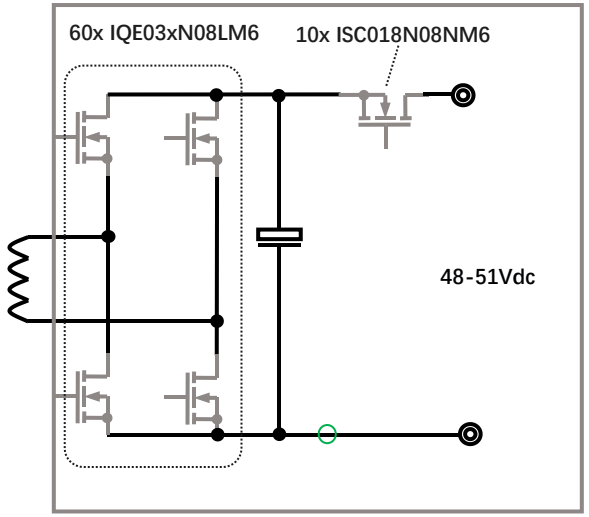
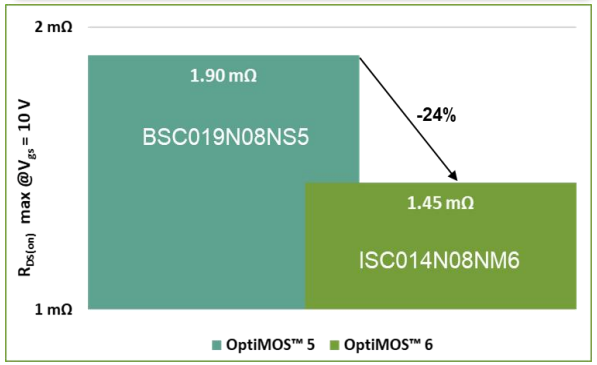
OptiMOS™ 6，兼顾导通损耗与开关损耗，提升全负载范围的效率

全桥同步整流+Or-ing

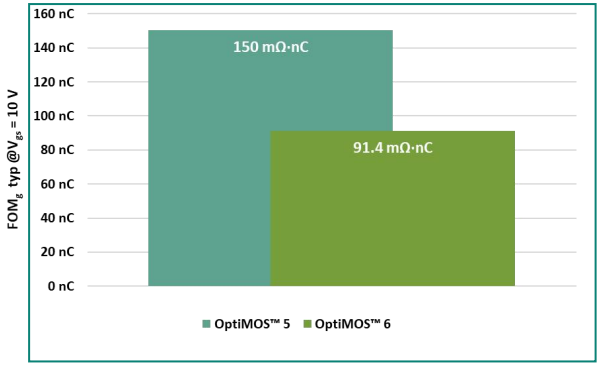


以80V PQFN 5*6封装为例

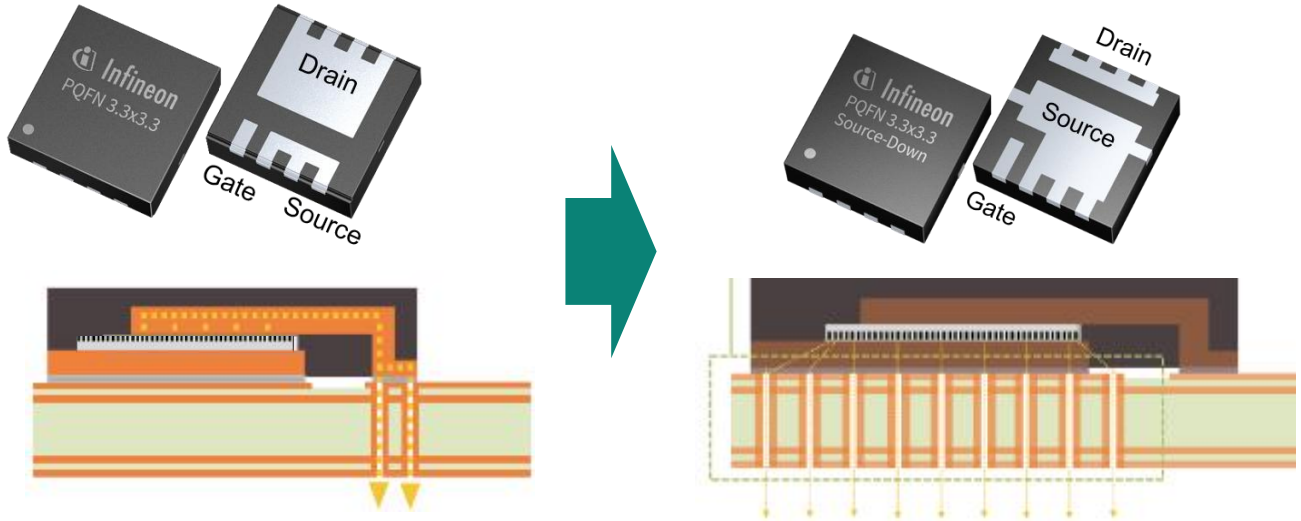
最低Rds(on) 降低24%



品质因数FOM_g 降低37%



创新的source down封装，更进一步提升热性能，助力高功率密度设计



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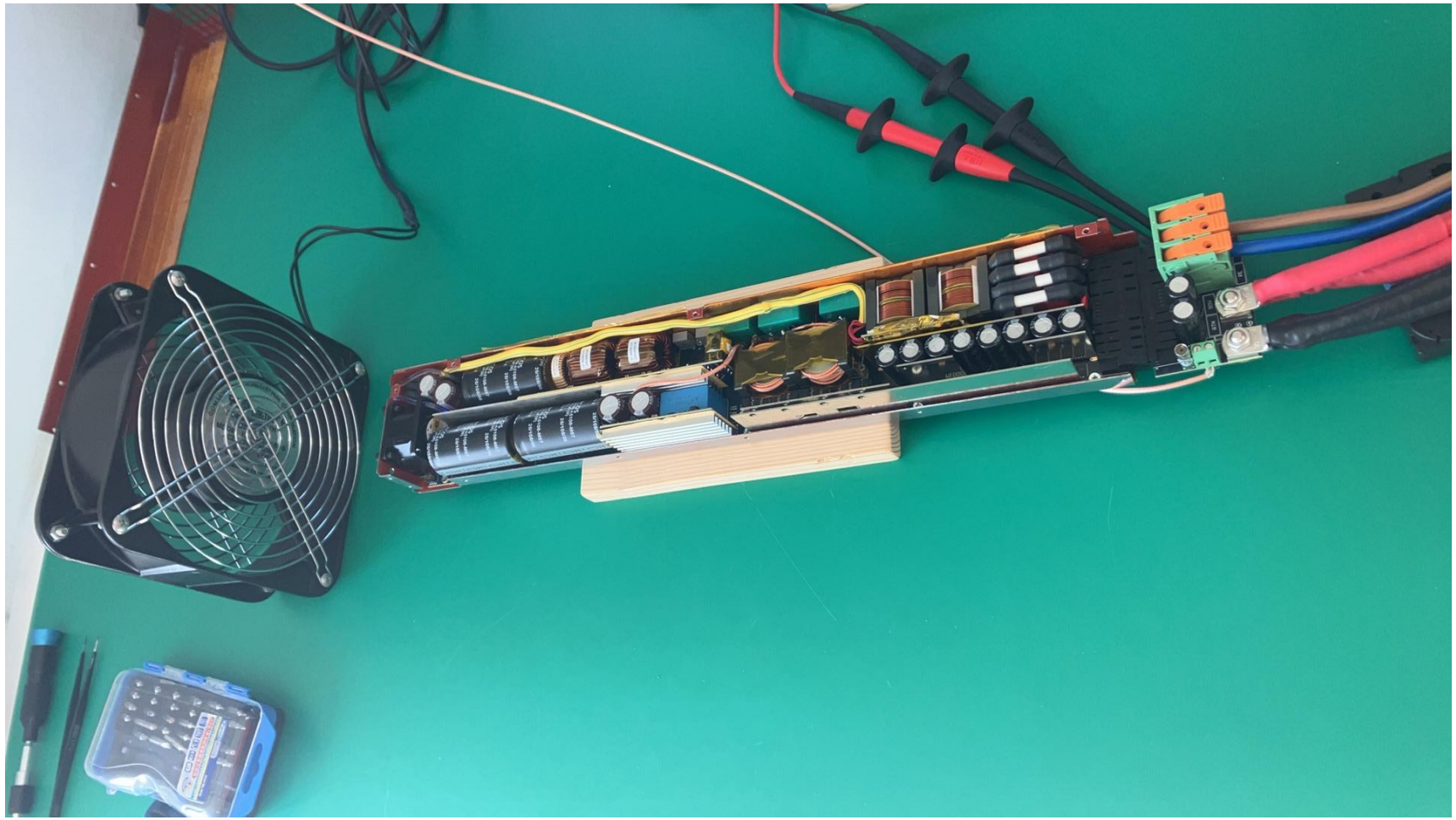
英飞凌（深圳）实验室，8KW参考设计开放测试

请联系接口的销售代表进行预约测试，英飞凌资深工程师现场提供详细讲解。



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